

Exercices on session *Statistics with R - Distributions and hypothesis*

Load the package tidyverse

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Admission in graduate school in UC Berkeley

We want to know if the admissions in graduate school in the Department A of UC Berkeley depends on the gender.

- The data are called *UCBAdmissions* and are available in base R. Load the data and look at their structure

```
data(UCBAdmissions)
str(UCBAdmissions)
```

```
'table' num [1:2, 1:2, 1:6] 512 313 89 19 353 207 17 8 120 205 ...
- attr(*, "dimnames")=List of 3
..$ Admit : chr [1:2] "Admitted" "Rejected"
..$ Gender: chr [1:2] "Male" "Female"
..$ Dept : chr [1:6] "A" "B" "C" "D" ...
```

- Get the data for department A (the first element of the array so you need to use the following command)

```
UCBAdmissions[, , 1]
```

	Gender	
Admit	Male	Female
Admitted	512	89
Rejected	313	19

```
dt_deptA <- UCBAdmissions[, , 1]
```

- These data give the number of admitted or rejected people per gender. How is this type of table called?

```
# a contingency table
```

- Perform a χ^2 test to see if the admission depends on the gender, and interpret it

```
res_chi2 <- chisq.test(dt_deptA)
res_chi2
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: dt_deptA
X-squared = 16.372, df = 1, p-value = 5.205e-05
```

```
# low p-value => yes the admission depends on the gender
```

- Check the expected frequency under the null hypothesis. Is the χ^2 test reliable?

```
res_chi2$expected
```

	Gender	
Admit	Male	Female
Admitted	531.4309	69.56913
Rejected	293.5691	38.43087

```
# yes reliable (all expected > 5)
```

- Get the residuals and interpret them

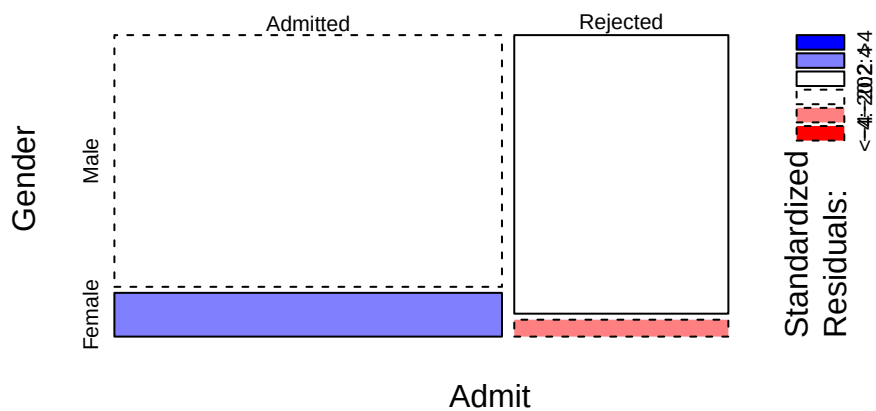
```
questionr::chisq.residuals(dt_deptA, std = TRUE)
```

	Gender	
Admit	Male	Female
Admitted	-4.15	4.15
Rejected	4.15	-4.15

```
# over-representation of the female in the admitted => female more admitted than male
```

- Represent the residuals graphically

```
mosaicplot(dt_deptA,
  shade = TRUE,
  main = "") # remove main title
```



Weight of cabbages from different cultivars

We want to know if the weight of cabbages planted on a certain day depends on the cultivar.

- Load the dataset called *cabbages* in the package *MASS*

```
library(MASS)
```

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

```
select
```

```
data("cabbages")
str(cabbages)
```

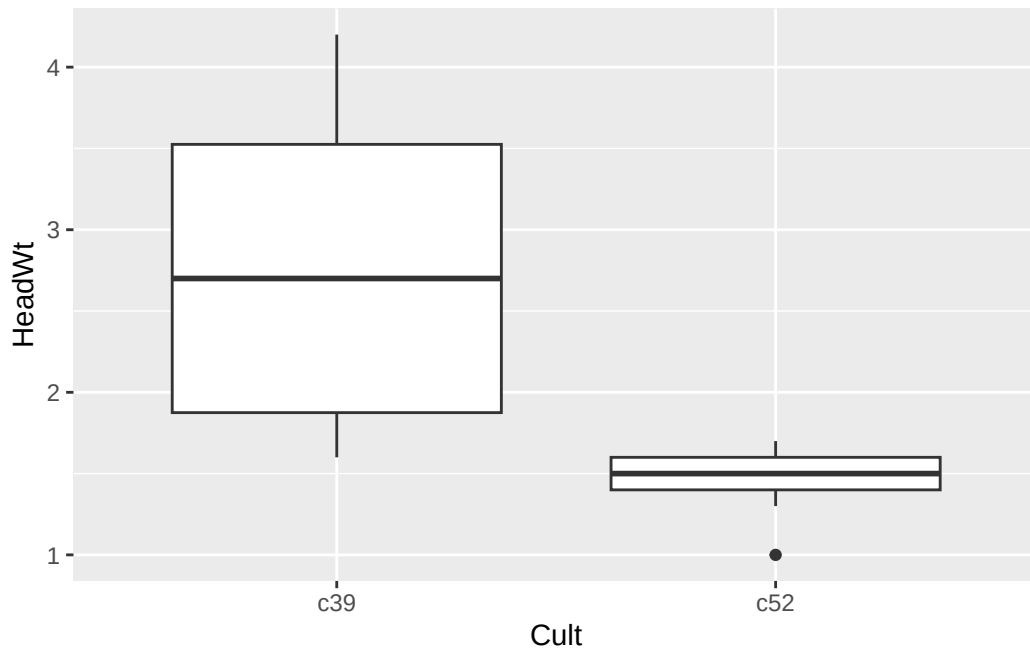
```
'data.frame': 60 obs. of 4 variables:
 $ Cult : Factor w/ 2 levels "c39","c52": 1 1 1 1 1 1 1 1 1 1 ...
 $ Date : Factor w/ 3 levels "d16","d20","d21": 1 1 1 1 1 1 1 1 1 1 ...
 $ HeadWt: num 2.5 2.2 3.1 4.3 2.5 4.3 3.8 4.3 1.7 3.1 ...
 $ VitC : int 51 55 45 42 53 50 50 52 56 49 ...
```

- Make a dataset containing only the cabbages planted at date *d21*

```
dt_cab <- cabbages %>% filter(Date == "d21")
```

- Make a boxplot to graphically explore if the weight depends on the cultivar. What do you think?

```
dt_cab %>%
  ggplot(aes(x = Cult, y = HeadWt)) +
  geom_boxplot()
```



- Check the normality of the distribution, using a test and graphically

```
# make a dataset per cultivar
dt_c39 <- dt_cab %>% filter(Cult == "c39")
dt_c52 <- dt_cab %>% filter(Cult == "c52")
```

```
# Shapiro test of normality (H0: the data are normally distributed)
shapiro.test(dt_c39$HeadWt)
```

Shapiro-Wilk normality test

```
data: dt_c39$HeadWt
W = 0.89345, p-value = 0.1854
```

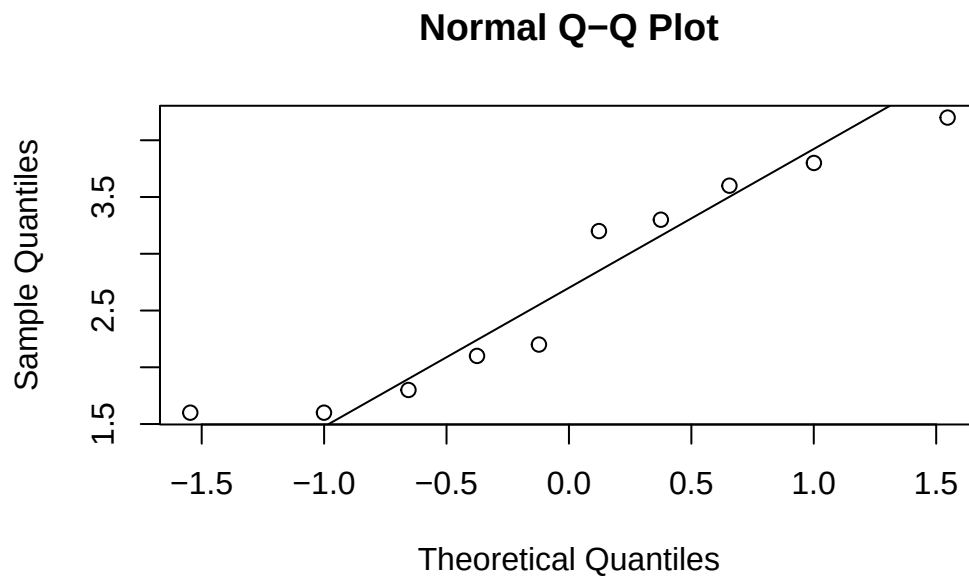
```
shapiro.test(dt_c52$HeadWt)
```

Shapiro-Wilk normality test

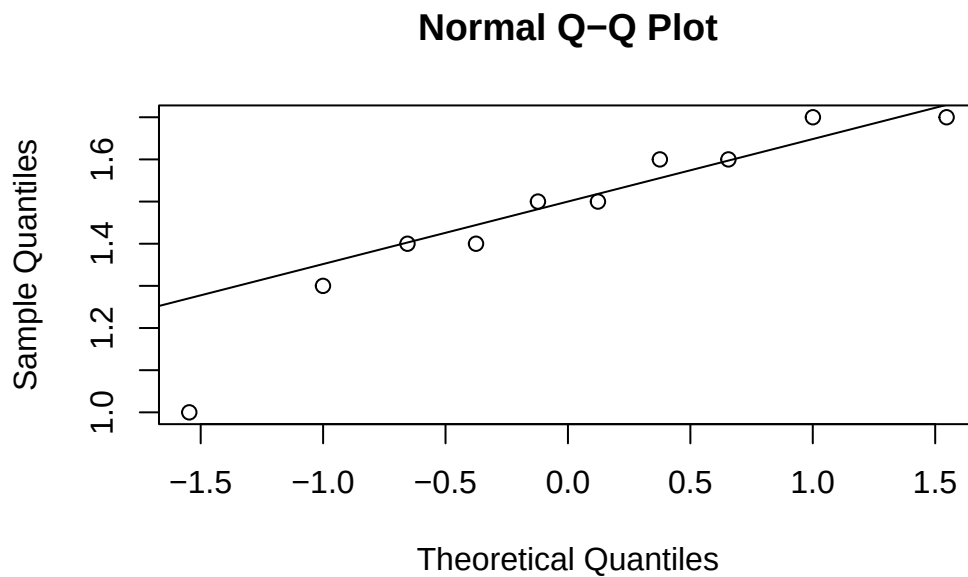
```
data: dt_c52$HeadWt
W = 0.8996, p-value = 0.2169
```

```
# not significant so the data are normally distributed
```

```
# Q-Q plots  
qqnorm(dt_c39$HeadWt)  
qqline(dt_c39$HeadWt)
```



```
# Q-Q plots  
qqnorm(dt_c52$HeadWt)  
qqline(dt_c52$HeadWt)
```



```
# very little points, so hard to interpret...
```

- Perform a t-test

```
res_ttest <- t.test(dt_cab$HeadWt ~ dt_cab$Cult)
res_ttest
```

Welch Two Sample t-test

data: dt_cab\$HeadWt by dt_cab\$Cult

t = 3.9929, df = 9.8275, p-value = 0.002637

alternative hypothesis: true difference in means between group c39 and group c52 is not equal

95 percent confidence interval:

0.5596128 1.9803872

sample estimates:

mean in group c39 mean in group c52

2.74 1.47

- Get the p-value, the mean estimated difference between the two groups and its confidence interval. What do you conclude?

```
res_ttest$p.value
```

```
[1] 0.002637016
```

```
# The difference is significant
```

```
cab_est <- res_ttest$estimate  
abs(cab_est[1] - cab_est[2])
```

```
mean in group c39  
1.27
```

```
# or  
abs(cab_est[2] - cab_est[1])
```

```
mean in group c52  
1.27
```

```
# The difference between group is big (compared to the mean per group)  
# Note that the name of the vector of result is not correct
```

```
res_ttest$conf.int
```

```
[1] 0.5596128 1.9803872  
attr("conf.level")  
[1] 0.95
```

```
# 0.95 is the confidence level  
# big confidence interval (few data points...)
```

Effect of the planting date on the content of vitamin C in cabbages

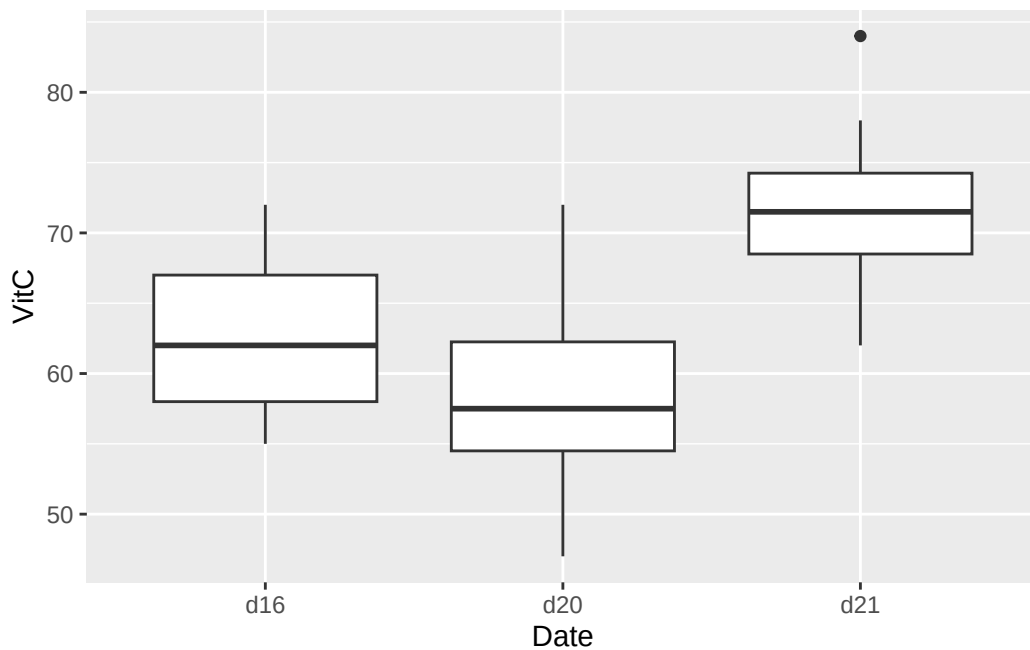
We want to know if the content of vitamin C in cabbages of the cultivar c52 depends on the planting date.

- Make a dataset containing only the cabbages of the cultivar *c52*

```
dt_cab2 <- cabbages %>% filter(Cult == "c52")
```

- Make a boxplot to explore the question graphically. What do you think?

```
dt_cab2 %>%
  ggplot(aes(x = Date, y = VitC)) +
  geom_boxplot()
```



- Perform an ANOVA

```
aov_cab <- aov(VitC ~ Date, data = dt_cab2)
summary(aov_cab)
```

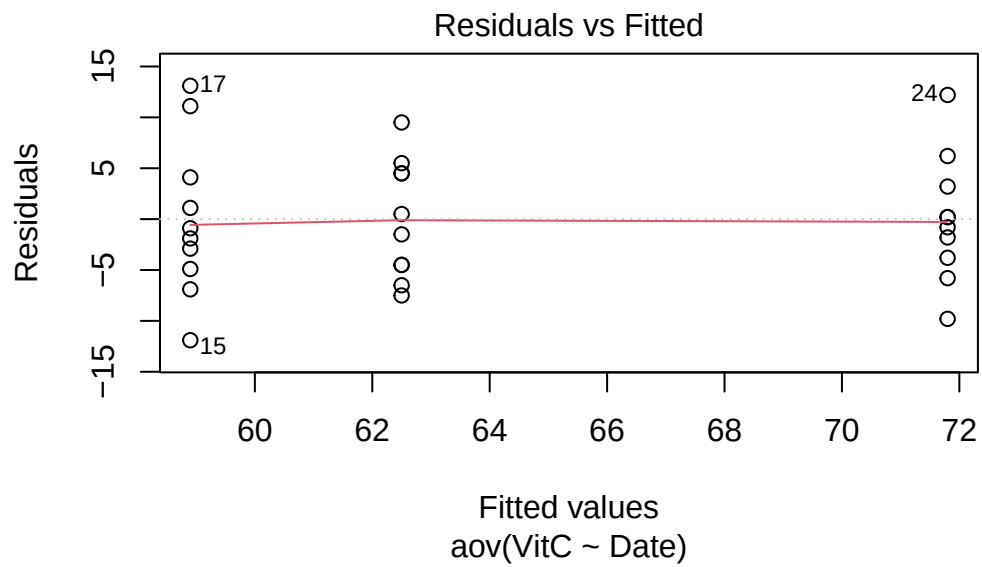
```
          Df Sum Sq Mean Sq F value    Pr(>F)
Date         2   886.2    443.1    10.08 0.000538 ***
Residuals   27  1187.0     44.0
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# the test is significant => the content of vitamin C depend on the planting date
```

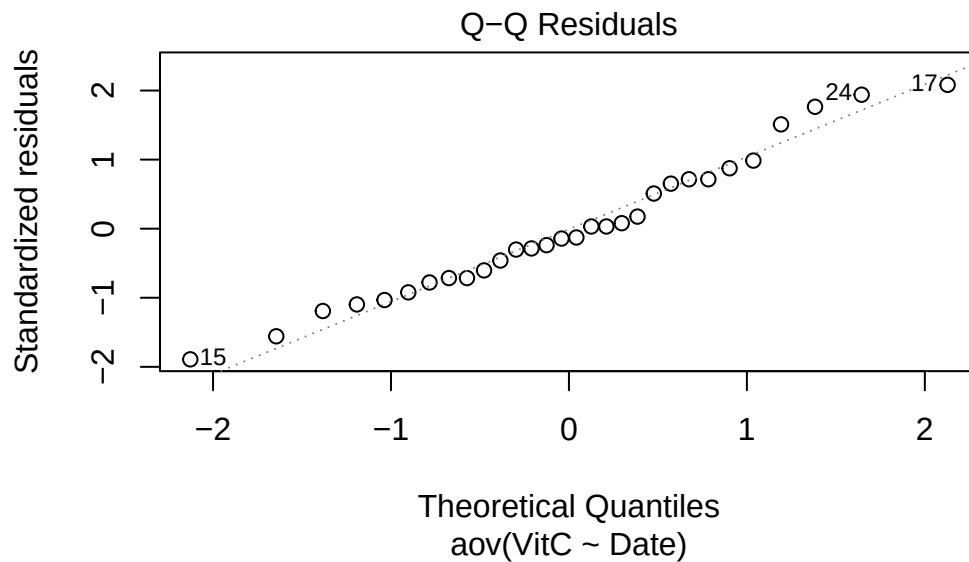
- Check the model assumptions

```
# check homoscedasticity
plot(aov_cab, which = 1)
```



```
# => OK
```

```
# check normality of the residuals
plot(aov_cab, which = 2)
```



```
# => OK
```

- Get the estimated mean of each groups

```
coef(aov_cab)
```

```
(Intercept)      Dated20      Dated21
        62.5         -3.6         9.3
```

```
# d16
```

```
coef(aov_cab)[1]
```

```
(Intercept)
        62.5
```

```
# d20
```

```
coef(aov_cab)[1] + coef(aov_cab)[2]
```

```
(Intercept)
        58.9
```

```
# d21
coef(aov_cab)[1] + coef(aov_cab)[3]
```

```
(Intercept)
      71.8
```

- Perform a Tukey test see which groups are different, and interpret it

```
tukey_cab <- TukeyHSD(aov_cab)
tukey_cab
```

```
Tukey multiple comparisons of means
 95% family-wise confidence level
```

```
Fit: aov(formula = VitC ~ Date, data = dt_cab2)
```

```
$Date
      diff      lwr      upr      p adj
d20-d16 -3.6 -10.952045  3.752045 0.4553220
d21-d16  9.3  1.947955 16.652045 0.0110721
d21-d20 12.9  5.547955 20.252045 0.0004962
```

```
# d21 differs from the two others
```

- Represent the results of the ANOVA graphically

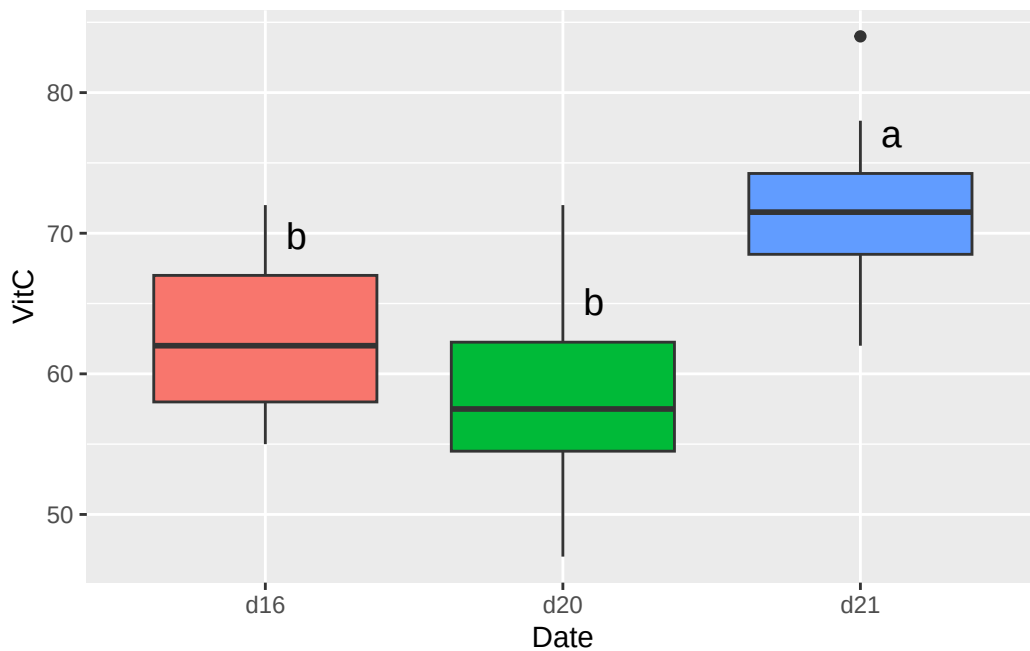
```
# get the letters
cab_let <- multcompView::multcompLetters4(aov_cab, tukey_cab)

# put them in a tibble
cab_let_tb <- tibble(
  Date = names(cab_let$Date$Letters),
  letter = as.character(cab_let$Date$Letters)
)

# make a tibble with the dates, the letters and the 75th percentile
cab_let_pos <- dt_cab2 %>%
  group_by(Date) %>%
  summarise(q_75 = quantile(VitC, probs = 0.75, na.rm = TRUE)) %>%
  inner_join(cab_let_tb)
```

Joining with ``by = join_by(Date)``

```
ggplot() +  
  geom_boxplot(data = dt_cab2,  
               aes(x = Date, y = VitC, fill = Date),  
               show.legend = FALSE) +  
  geom_text(data = cab_let_pos,  
            aes(x = Date, y = q_75, # position of the label  
                label = letter), # labels are the letters  
            size = 5, # size of the labels  
            vjust = -1, hjust = -1) # adjusts position of the labels
```



Brain and body weights in mammals

We want to know if there is a correlation between the brain and the body weights in mammals.

- Load the dataset called *mammals* in the package *MASS*

```
data(mammals)  
str(mammals)
```

```
'data.frame': 62 obs. of 2 variables:  
 $ body : num 3.38 0.48 1.35 465 36.33 ...  
 $ brain: num 44.5 15.5 8.1 423 119.5 ...
```

- Get the Pearson correlation coefficient

```
cor(mammals$body, mammals$brain)
```

```
[1] 0.9341638
```

```
# very high correlation
```

- Do a Pearson correlation test and interpret it

```
cor.test(mammals$body, mammals$brain,  
         method = "pearson")
```

Pearson's product-moment correlation

```
data: mammals$body and mammals$brain  
t = 20.278, df = 60, p-value < 2.2e-16  
alternative hypothesis: true correlation is not equal to 0  
95 percent confidence interval:  
 0.8926796 0.9599518  
sample estimates:  
 cor  
0.9341638
```

```
# There is a strong significant positive relationship between body and brain mass
```